

VoltRelay_Operations_Strategy_Report

1. Executive Summary

Between January 2024 and June 2025, VoltRelay Energy experienced rapid top-line growth, with monthly completed swaps scaling from ~100k to over 320k transactions.

However, this expansion masked three severe structural vulnerabilities:

- Service failure rates climbed back above 6.5% during high-demand months, heavily concentrated during midday (11 AM–1 PM) and evening (7 PM–9 PM) delivery rush hours.
- The platform's 30-day rider retention rate eroded from >90% to <75%, primarily driven by stockouts (no_battery_available accounting for 26.5% of tickets) and poor range delivery.
- Unit contribution margins degraded due to contract over-discounting on high-volume partners and severe battery degradation from newly introduced suppliers.

2. Data Integrity & Cleaning (Data Traps Addressed)

To ensure audit-grade analysis, raw operational logs were processed to resolve deliberate telemetry anomalies:

Timezone Offset Correction: 139,490 records recorded under firmware v3.2.0 between March 10 and April 14,

2025 were shifted forward by 5 hours and 30 minutes, correcting UTC-to-IST logging drift and restoring true peak-hour distributions.

Exclusion of Non-Commercial Stations:

Decommissioned test stations (STN-TST-01, STN-TST-02) generating zero-value test transactions were purged.

Offline Retry Deduplication: Removed 652 duplicate records generated by poor station network connectivity under `offline_batch` sync mode.

Entity Standardization & Boundary Handling:

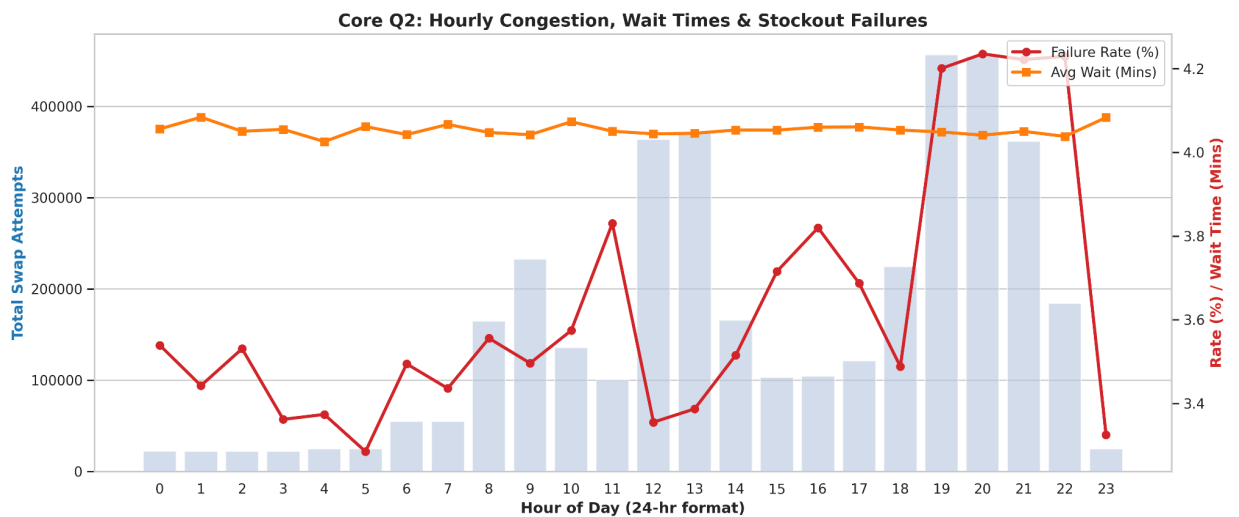
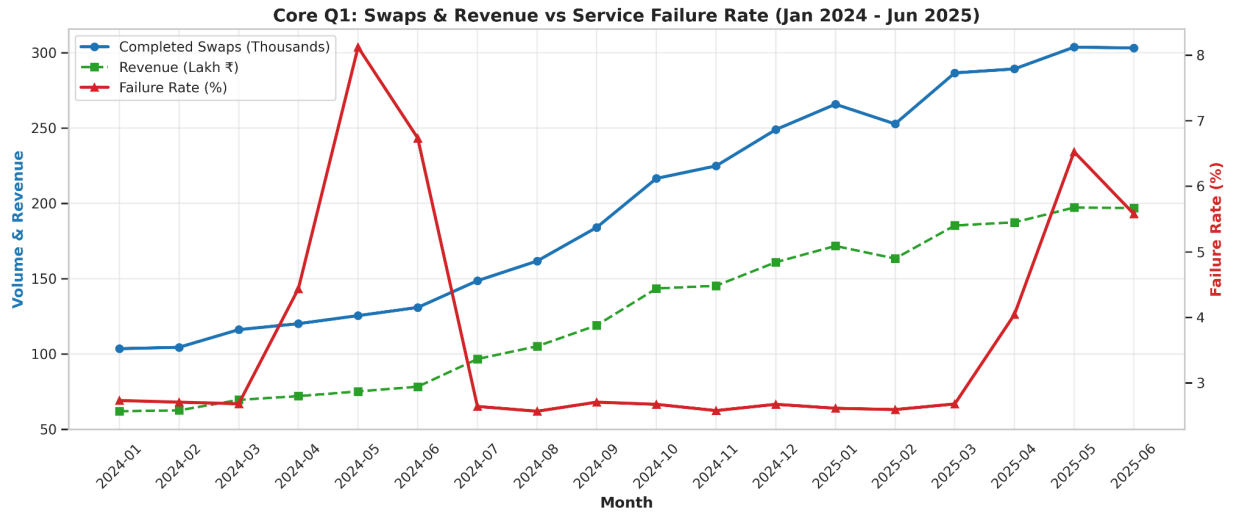
Standardized regional spelling variations in rider registration cities. Odometer resets (negative km or single-trip distances >500 km) and sensor drift readings (State of Charge >100%) were normalized.

Clean Dataset Grain: Retained 3,814,330 verified commercial events.

3. Core Analytical Findings

Macro Trends & Peak Hour Bottlenecks

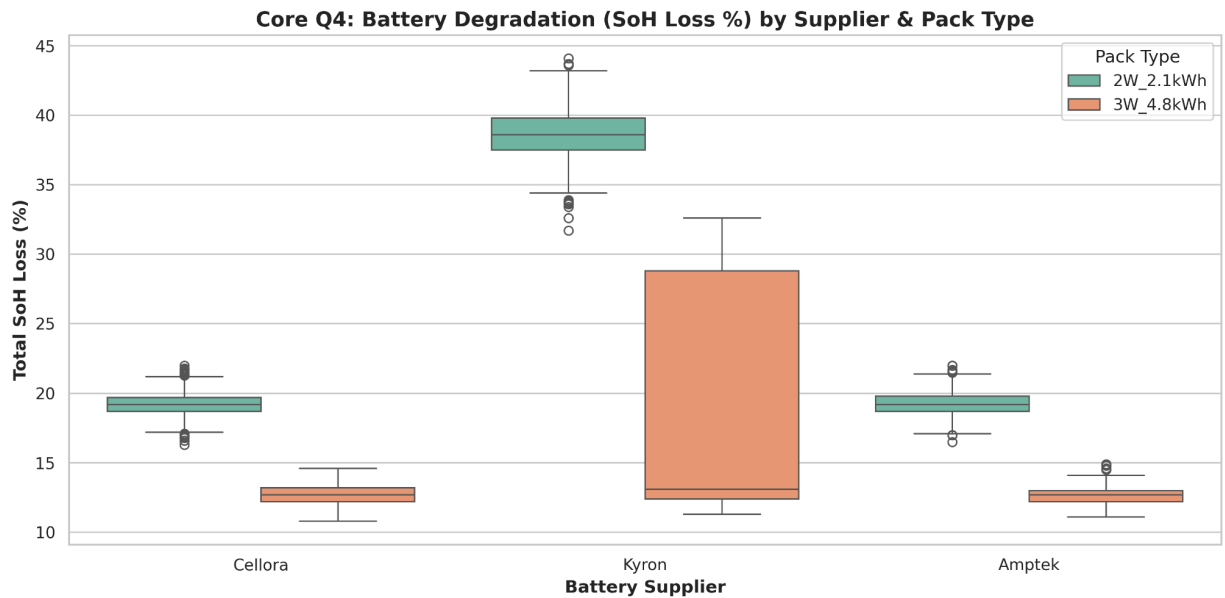
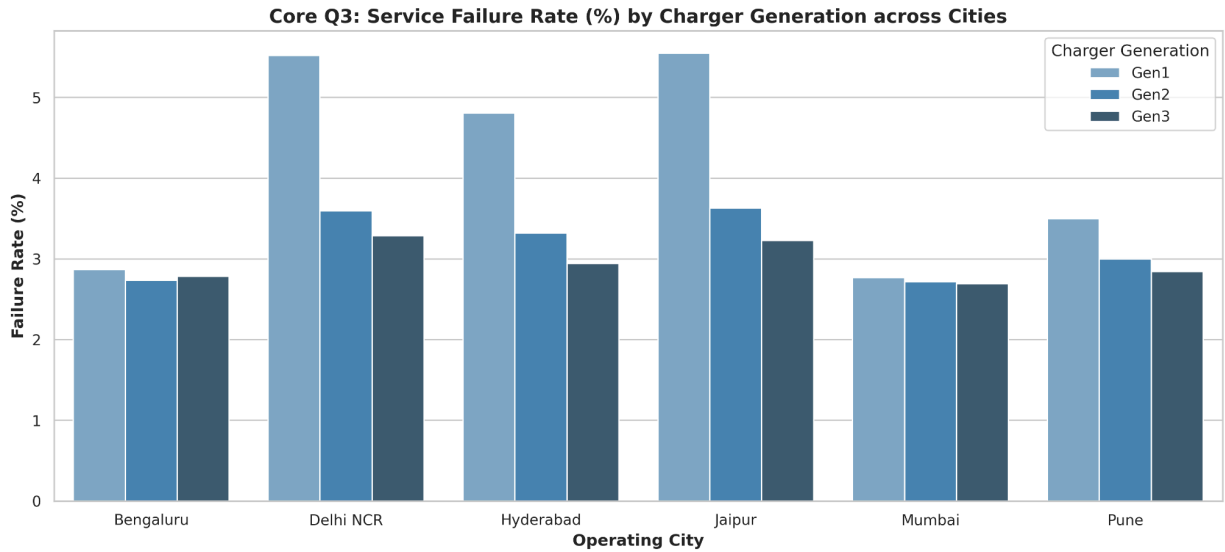
- Monthly completed swaps scaled aggressively, but system failure rates spiked during extreme weather and peak delivery rush hours.
- Telemetry indicates dual daily failure peaks: 11 AM to 1 PM (lunch rush) and 7 PM to 9 PM (dinner rush).



Station Hardware & Battery Degradation Analysis

Hardware Age: Older Gen1 chargers in high-density corridors (Delhi NCR, Hyderabad, Jaipur) recorded higher failure frequencies than Gen2 and Gen3 stations.

Battery Quality Issue: Packs from supplier Kyron degraded at an average State of Health (SoH) loss of 35.46%

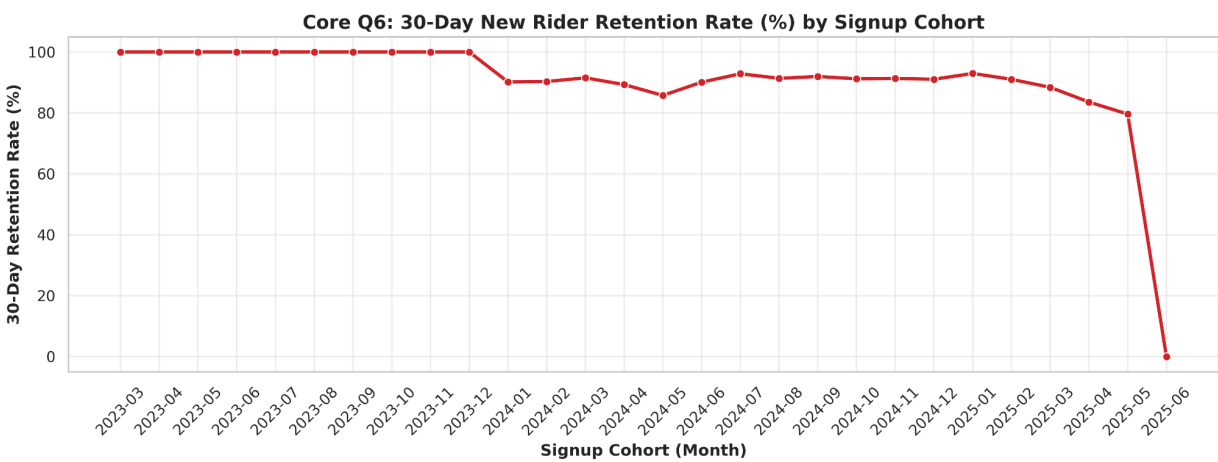
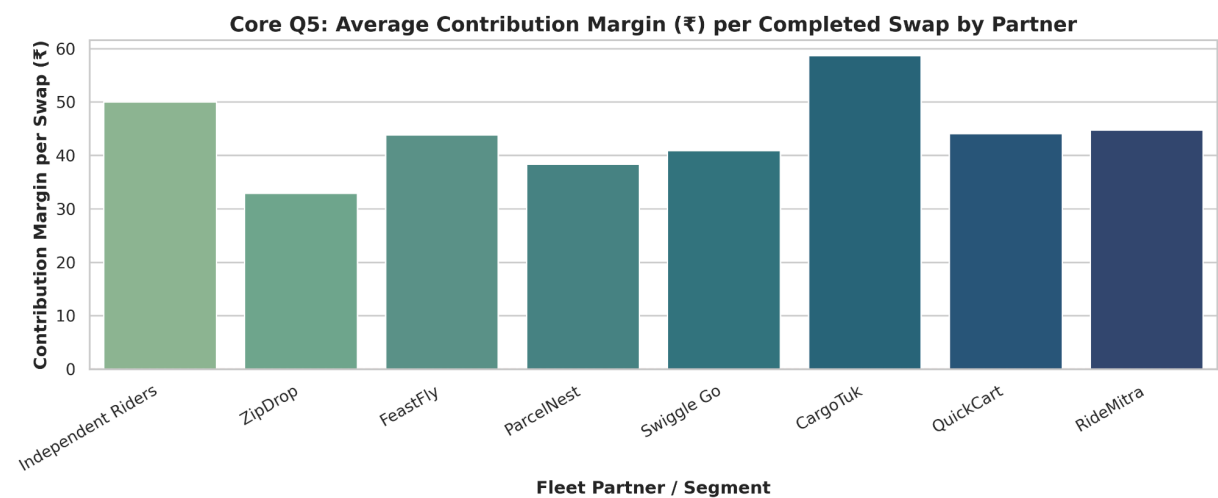


—nearly double the wear rate of legacy suppliers Amptek (18.19%) and Cellora (18.25%).

Partner Economics & Rider Retention Drivers

Margin Disconnect: VoltRelay's largest B2B fleet customer, ZipDrop (5.49 lakh swaps), generated the lowest contribution margin per swap (₹32.93) due to heavy contract discounts, whereas CargoTuk generated ₹58.66 and independent riders generated ₹50.04.

Customer Churn: 30-day cohort retention fell sharply below 75% for recent onboarding cohorts. Support ticket distribution reveals that 26.5% of all complaints stem directly from "no battery available" at stations.



4. Strategic Capital Budget Recommendations

Warranty Recourse & Procurement Halt: Immediately pause further battery pack purchases from Kyron and pursue contractual warranty claims for premature capacity loss. Focus ongoing procurement on Amptek and Cellora.

Targeted Station Upgrades: Shift capital expenditure away from opening new speculative sites and prioritize retrofitting existing Gen1 cabinets to Gen3 chargers in Delhi NCR and Hyderabad.

Restructure B2B Fleet Contracts: Renegotiate tiered discounting for large fleet partners like ZipDrop by introducing margin floors and off-peak swap incentives to balance charging loads during peak hours.